

GridPilot: A Convex Optimization Framework for Carbon-Intelligent Electric Vehicle Fleet Charging in Indian Corporate Depots

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Abstract

India's transition to electric vehicles (EVs) is frequently justified on climate grounds, yet the carbon benefit of an EV depends entirely on when it draws power from a grid whose emission intensity varies through the day. Uncoordinated overnight charging at large corporate EV depots also risks transformer overloads and inflates demand charges under Indian time-of-use tariff structures, costs typically passed through to fleet operators and, indirectly, to the workers and passengers those fleets serve. This paper presents GridPilot, a convex quadratic programming (QP) framework that jointly minimizes grid carbon intensity, transformer peak load, vehicle charging discomfort, and utility demand charges for a simulated 600-vehicle corporate depot in Gurugram, Haryana. Using CVXPY with the CLARABEL interior-point solver, the model produces a mathematically optimal, globally guaranteed schedule across 57,600 decision variables (600 vehicles by 96 fifteen-minute slots). Relative to unmanaged charging, the optimized schedule reduces peak transformer load by 55.1% (4,456 kW to 2,000 kW), eliminates three nightly transformer overload events, saves an estimated ₹8.60 lakh per month in demand charges under the Dakshinanchal Vidyut Vitran Nigam Limited (DVVNL) HT-2 tariff, and avoids 2,072 kg of CO₂ per night by shifting load into a period of lower grid carbon intensity, while still delivering all 600 vehicles to 80% state of charge by

07:00 IST. We situate these results within the climate justice literature on energy access and least-cost decarbonization pathways, and discuss the framework's applicability to India's broader corporate EV fleet sector.

Keywords: electric vehicle charging, convex optimization, demand-side management, carbon intensity, climate justice, grid decarbonization, India

1. Introduction

Electric vehicles are widely presented as a low-carbon substitute for internal combustion transport, and Indian corporate fleet operators have begun large-scale EV adoption for last-mile and staff transport in cities such as Gurugram, Bengaluru, and Pune. This substitution is not automatically climate-positive, however, because the emissions attributable to an EV depend on the carbon intensity of the electricity used to charge it. India's grid remains substantially coal-dependent, and its emission factor varies meaningfully across the day: it is highest during the early-evening peak, when thermal and gas plants must cover demand that renewable generation cannot meet, and lowest in the pre-dawn hours, when demand is low relative to available baseload and renewable capacity. A depot that charges its entire fleet immediately upon return, typically between 20:00 and 22:00 IST, charges during the dirtiest part of the day, eroding much of the climate advantage the EVs were adopted to capture.

The same uncoordinated charging pattern creates a second, more immediate problem for fleet operators: distribution transformer overload. Corporate depots are typically served by a fixed transformer capacity sized for existing building loads, not for a fleet of dozens or hundreds of vehicles charging simultaneously. When charging demand exceeds this capacity, operators face equipment stress, safety risk, and demand charges levied under India's maximum-demand tariff structures, which bill on the single highest 15-minute average demand recorded in a billing cycle

rather than on total energy consumed. A fleet operator who could shift charging in time, without changing total energy delivered, could avoid both the emissions penalty and the financial penalty simultaneously.

This is fundamentally a scheduling and resource-allocation problem, and it admits a rigorous mathematical treatment. This paper models a corporate EV depot as a convex quadratic program: a per-vehicle, per-timeslot charging power schedule chosen to minimize a weighted combination of carbon emissions, transformer peak load, vehicle charging discomfort, and demand-charge exposure, subject to hard constraints on charger capacity, vehicle availability windows, and next-morning readiness. Because the problem is convex, the solution returned is provably globally optimal, not merely a locally good heuristic outcome, which distinguishes this approach from rule-based or proportional load-balancing schemes already deployed by some Indian operators.

We frame this work explicitly within a climate justice lens. Climate justice scholarship emphasizes that the costs and benefits of decarbonization are unevenly distributed, and that mitigation strategies should be evaluated not only for aggregate emissions reduction but for who bears the transition costs, including energy costs passed on to workers, commuters, and low-margin fleet operators. An EV fleet transition that inadvertently produces high electricity bills, transformer damage, or precarious grid conditions in the depot's neighbourhood displaces the burden of decarbonization onto exactly the actors least able to absorb it. A scheduling system that reduces both carbon intensity and financial cost, without added hardware, offers a route to decarbonization that does not create new distributional harms.

The remainder of this paper is organized as follows. Section 2 reviews related work in EV charging optimization and situates the gap this paper addresses. Section 3 describes the reference depot and data sources. Section 4 presents the mathematical formulation. Section 5 reports simulation results.

Section 6 discusses the findings through a climate justice framing. Section 7 addresses limitations, and Section 8 concludes.

2. Literature Review

A substantial body of work addresses EV charging coordination, though most of it targets residential or public charging contexts rather than captive corporate depots, and comparatively little of it engages with Indian tariff structures or grid carbon-intensity data. Reinforcement learning approaches have been proposed for online EV charging scheduling in settings where future arrivals and departures are unknown in advance, formulating the problem as a cost-and-peak-minimization task solved without a forecast of future vehicle behaviour (Cao, Wang, Li, & Zhang, 2021). Such model-free methods trade the guarantee of global optimality for adaptability to uncertain, streaming information, which is valuable where vehicle behaviour is highly stochastic but less necessary in a corporate depot where fleet size, arrival windows, and energy needs are comparatively predictable and knowable in advance.

Peak-constrained scheduling has also been studied in adjacent infrastructure contexts. An energy management system for EV charging at electric railway stations incorporates peak load constraints set by the grid operator directly into the optimization, treating each vehicle's state of charge at departure as a decision variable to be maximized subject to those constraints (Pierrou, Valero-De la Flor, & Hug, 2024). This is structurally close to the corporate depot problem this paper addresses, though it does not incorporate a time-varying carbon-intensity signal or a demand-charge-style financial objective, both of which are central to the Indian tariff and grid context.

A parallel literature addresses transformer-level impacts specifically. Fuzzy-logic and Monte Carlo-based charging controllers have been shown to reduce distribution transformer peak loading and improve voltage stability under rising EV penetration, using transformer load level itself,

together with vehicle state of charge, as the controlling signal for charging power. Reviews of the broader smart-charging and vehicle-grid-integration literature likewise emphasize that demand response strategies which shift EV load to off-peak hours reduce grid congestion and cost, and that centralized, cloud-based coordination architectures allow real-time adjustment of charging schedules to grid conditions. These strands establish transformer protection and cost reduction as well-studied objectives, but they are typically pursued independently, using heuristic or metaheuristic solvers (genetic algorithms, fuzzy logic, swarm optimization) rather than a convex formulation with global optimality guarantees, and they rarely combine an explicit carbon-intensity objective with an explicit utility demand-charge objective in a single formulation calibrated to a specific national grid and tariff.

India-specific work on EV smart charging remains comparatively sparse in the peer-reviewed literature, and existing commercial dynamic load management products used by Indian fleet operators, such as those offered by domestic charging-infrastructure vendors, generally distribute available power proportionally across connected vehicles in response to real-time constraints rather than solving a full-night, per-vehicle optimization in advance. This is a materially different problem: proportional distribution reacts to constraint violations as they occur, whereas an advance per-vehicle QP schedule anticipates constraints and allocates power to avoid them, at the cost of assuming that vehicle arrival and energy-need information is known reasonably far ahead, which is realistic for a captive corporate depot with a known daily fleet return pattern. This paper's contribution is to formulate that problem as a convex QP calibrated jointly to Indian grid carbon-intensity data, an Indian corporate EV fleet composition, and an Indian time-of-use demand-charge tariff, and to report simulated results at a scale representative of a large Indian fleet operator.

2.1 Commercial Load Management Tools in India

Commercial EV charging platforms operating in the Indian market, including white-label charge point management providers such as Evoltsoft, address the transformer-capacity problem through dynamic load management: a controller continuously monitors total site draw and throttles individual chargers in real time so that combined demand stays within the site's electrical headroom, without requiring a physical grid upgrade (Evoltsoft, n.d.). This is a genuinely useful and widely deployed capability, and it solves a real operational problem. It is, however, a reactive mechanism: power is redistributed after vehicles are already connected and drawing current, in response to the constraint as it is encountered, rather than planned in advance of it. None of the commercial dynamic load management products reviewed for this paper incorporate a time-varying grid carbon-intensity signal or an Indian utility demand-charge structure into their allocation logic; their objective is constraint satisfaction at the site level, not joint minimization of carbon impact and billing cost across a full charging cycle.

GridPilot addresses a different version of the same underlying problem. Because a corporate depot's fleet size, return time, and next-morning energy requirement are known well in advance of the charging window, the scheduling decision does not need to be made reactively. Computing the full night's per-vehicle schedule ahead of time, rather than adjusting it moment to moment, allows the carbon-intensity and demand-charge signals to be built directly into the objective rather than treated as an afterthought layered on top of a capacity constraint.

2.2 Positioning and Novelty of GridPilot

It is worth being precise about what is and is not novel here, since convex and quadratic programming approaches to EV charging are not new to the literature. A quadratic programming formulation for electric bus fleet charging has previously reported an average peak load reduction of 83%, notably larger than the 55.1% reported in this paper (Houbbadi, Trigui, Pelissier, Redondo-

Iglesias, & Bouton, 2019). That comparison is worth confronting directly rather than avoiding: the bus fleet study models vehicle-to-grid discharge as an available lever, which gives its optimizer a degree of freedom, sending power back to the grid during peak hours, that a one-directional corporate charging depot does not have. Reinforcement learning formulations of EV charging scheduling likewise exist, but they are typically designed for settings where future arrivals are unknown and trade the guarantee of a globally optimal solution for the ability to adapt online. Peak-constrained convex formulations have been applied to other fixed-infrastructure contexts, such as EV charging at railway station car parks, but without an explicit carbon-intensity term.

What appears genuinely underexplored, at least in the sources reviewed for this paper, is the specific combination GridPilot implements: a single convex program that jointly minimizes grid carbon intensity, transformer peak load, per-vehicle charging discomfort, and utility demand-charge exposure, solved exactly rather than approximated, and calibrated to Indian fleet composition, Indian grid carbon-intensity data, and an Indian time-of-use tariff at the scale of an actual corporate fleet operator. The mathematical technique, convex quadratic programming, is established. The calibration and the four-term joint objective, applied together at this scale and in this regulatory context, is the paper's actual contribution, and it should be described that way rather than as a claim that no comparable optimization approach exists anywhere in the literature. It is also worth noting that, in most Indian grids, the evening period of highest carbon intensity and the evening period in which utilities record peak billed demand coincide for structural reasons: both reflect the same scarcity of low-cost generation capacity at that hour. A schedule optimized primarily for one objective is therefore likely to move substantially toward the other regardless; the value of the explicit joint formulation is that it guarantees neither objective is sacrificed to over-optimize the other, and that the resulting savings on both fronts can be quantified and reported rather than assumed as a side effect.

2.3 Competitive Position

Among Indian EV charging software providers reviewed for this paper, none was found to combine a per-vehicle convex optimizer, Indian government fleet and grid data, and physical power-flow validation in a single system. Evoltsoft's dynamic load management distributes available power across connected chargers reactively, in response to real-time site conditions, which is a different computational problem from solving a full-night, per-vehicle schedule in advance (Evoltsoft, n.d.). Other Indian and India-facing charging management platforms, including Pulse Energy, Numocity, and ChargePoint's Indian operations, offer charger monitoring, billing, and load-balancing features but were not found to publish a per-vehicle convex optimization layer validated against AC power-flow physics, which GridPilot performs using a seven-bus pandapower model of the depot's internal distribution network. This distinction, reactive proportional allocation versus proactive per-vehicle optimization with physical validation, is the basis for GridPilot's competitive position rather than a claim of being the only optimization-based system in the market.

3. Reference Depot and Data Sources

The reference case is a corporate EV depot in Gurugram, Haryana, modelled on the scale and vehicle mix of a large Indian fleet operator. The depot is sized at 600 vehicles, drawn from the 2025 national vehicle registry composition for corporate EV fleets, comprising six models across three manufacturers: Tata Motors (Nexon EV, Tiago EV, Curvv EV), JSW MG Motor (Windsor EV, ZS EV), and Mahindra (BE6). Table 1 summarizes the fleet composition used in the simulation.

Model	Count	Share	Battery (kWh)	Charger (kW)
Tata Nexon EV	162	27%	30.2	7.4
Tata Tiago EV	84	14%	19.2	3.3

MG Windsor EV	108	18%	38.0	7.4
Mahindra BE6	102	17%	59.0	7.2
Tata Curvv EV	96	16%	55.0	7.2
MG ZS EV	48	8%	50.3	7.4
Total	600	100%		

Table 1. Reference fleet composition, adapted from Vahan Dashboard (Ministry of Road Transport and Highways)

2025 registration data and EVreporter market-share analysis.

The weighted-average vehicle in this fleet carries a 40.53 kWh battery and a 6.75 kW onboard charger. Each vehicle is modelled as requiring energy sufficient to move from an assumed 20% arrival state of charge to an 80% departure state of charge, yielding 24.32 kWh per vehicle per session and a total nightly energy requirement of 14,592 kWh across the fleet. The 80% departure target, rather than 100%, reflects two considerations: lithium-ion cells entering the constant-voltage charging phase above roughly 80% state of charge draw power at a fraction of rated speed for comparatively little additional range, and repeated full-charge cycling accelerates battery degradation. Eighty percent state of charge is also judged sufficient for the 80-150 km daily duty cycles typical of Indian corporate fleet vehicles.

Charging is scheduled across a 20:00-07:00 IST overnight window, divided into 96 fifteen-minute timeslots. Fifteen-minute resolution is used because Indian time-of-use demand-charge tariffs, including the DVVNL HT-2 tariff used in this study, define maximum demand as the highest average load recorded over any fifteen-minute interval in the billing period; scheduling at this resolution allows the model's peak-load variable to map directly onto the utility's billing quantity. It is also the de facto granularity used by the Open Charge Point Protocol (OCPP) 1.6 SetChargingProfile command across major Indian charging hardware vendors, which supports eventual deployment to physical chargers.

Grid carbon intensity is drawn from the Central Electricity Authority's CO₂ Baseline Database (version 21, covering FY2024-25), which reports an annual average emission factor of 0.710 kg

CO₂ per kWh for the Haryana grid, with a lower rate of 0.631 kg CO₂/kWh during a clean overnight window (02:00-05:00 IST) and a higher rate of 0.771 kg CO₂/kWh during the early-evening dirty window (18:00-22:00 IST) when the fleet actually returns. Utility tariff parameters are drawn from the DVVNL HT-2 commercial tariff for FY2025-26, which levies a demand charge of ₹350 per kVA of billed maximum demand against a contracted demand of 5,000 kVA, with an additional ₹430 per kVA penalty for demand recorded above the contracted level. Vehicle arrival-time distribution shape, though not absolute energy or charger values, is adapted from the ACN-Data public charging dataset (Lee, Li, & Low, 2019), rescaled to the 20:00-22:00 Indian depot return window.

4. Methodology

4.1 Optimization Formulation

The scheduling problem is formulated as a convex quadratic program. The decision variable is $\text{power}[v, t]$, the charging power in kW delivered to vehicle v during timeslot t , for $v = 1$ to 600 and $t = 1$ to 96, giving 57,600 continuous, non-negative decision variables. The objective function is a weighted sum of four terms:

$$\text{minimize } \alpha \cdot C(x) + \beta \cdot P(x) + \gamma \cdot D(x) + \delta \cdot V(x)$$

where $C(x)$ is a carbon cost term (weight $\alpha = 0.50$) equal to the CEA grid-intensity-weighted energy delivered, which penalizes charging during the dirty window and rewards charging during the clean window; $P(x)$ is a peak penalty (weight $\beta = 0.20$) equal to the sum, across all timeslots, of the squared excess of total fleet load above a 2,000 kW target, a quadratic term that is what makes the problem a genuinely quadratic (rather than purely linear) program and which produces smooth load-spreading behaviour rather than sharp on/off cliffs at the target; $D(x)$ is a discomfort penalty

(weight $\gamma = 0.20$) equal to any shortfall between a vehicle's required energy and its delivered energy, which drives the model toward fully meeting every vehicle's 80% state-of-charge target; and $V(x)$ is a demand-charge backstop penalty (weight $\delta = 0.10$) that specifically penalizes peak load above the 4,500 kW DVVNL penalty-tier threshold, acting as a financial safety net on top of the general peak penalty. The four weights sum to 1.0.

The program is subject to four constraint families: charging power for every vehicle and timeslot must be non-negative; charging power in any slot cannot exceed the vehicle's onboard charger rating, and only during timeslots in which the vehicle is physically present and plugged in; each vehicle's total delivered energy across the night must reach at least 80% of its required energy, guaranteeing next-morning readiness; and total fleet load plus base building load in every timeslot must not exceed a 5,000 kW hard transformer safety cap, independent of the softer, weighted peak penalty in the objective.

The resulting convex QP is solved using CVXPY 1.4.1 as the modelling layer and the CLARABEL interior-point solver as the backend. CLARABEL was selected over commercial alternatives such as Gurobi or MOSEK for two reasons specific to this deployment context. First, both Gurobi's and MOSEK's free academic licenses require activation from an academic network, which is incompatible with a production system hosted on commercial cloud infrastructure. Second, and more materially for eventual commercialization, both vendors' academic licenses explicitly prohibit commercial use; a paid commercial Gurobi license costs an estimated \$10,000-16,000 per year, whereas CLARABEL is released under the Apache 2.0 license with no restriction on commercial deployment. Because the solver itself is not the dominant cost in the pipeline's total latency, this substitution carries negligible performance penalty relative to its licensing benefit.

4.2 System Architecture

The simulation pipeline comprises five layers. A data layer ingests the CEA carbon-intensity baseline, the national vehicle-registry fleet composition, the DVVNL tariff schedule, and weather data from the Open-Meteo API. A forecasting layer uses a Prophet time-series model (mean absolute percentage error of 0.83% on validation data) to project next-day vehicle arrival patterns, paired with an isolation-forest anomaly detector for irregular arrival behaviour. An optimization layer implements the QP described above and additionally validates the resulting schedule against a seven-bus AC power-flow model of the depot's internal distribution network using pandapower, confirming that the schedule which is optimal in the simplified kW-balance formulation does not introduce voltage violations in a more physically detailed representation; the unmanaged case produces fourteen voltage violations across the modelled feeders, all of which are eliminated under the optimized schedule. An API layer, built on FastAPI with a SQLAlchemy/SQLite persistence layer, exposes the computed schedule and stores run history. A dispatch layer communicates the schedule to individual chargers using the OCPP 1.6 SetChargingProfile command over WebSocket connections, the same standard used by major charging-hardware vendors operating in the Indian market.

5. Results

Table 2 summarizes the principal simulation outcomes comparing unmanaged (immediate, uncoordinated) charging against the GridPilot-optimized schedule for the 600-vehicle reference depot.

Metric	Unmanaged	GridPilot-Optimized
Peak transformer load	4,456 kW	2,000 kW
Transformer utilization	111.4%	50.0%
Overload events per night	3	0
DVVNL demand charge	~₹19.5L/month	~₹8.75L-₹10.9L/month

Estimated monthly saving	-	₹8.60 lakh
CO ₂ emitted per night	11,252 kg	9,208-9,180 kg
Estimated CO ₂ saved per night	-	2,072 kg
Vehicles ready by 07:00 IST	600/600	600/600
Decision variables	-	57,600
Solver time (CLARABEL)	-	~3,000 ms
Total API response (warm)	-	~5,500 ms

Table 2. Simulation results, 600-vehicle reference depot, single overnight cycle.

The optimized schedule reduces peak transformer demand by 55.1%, from 4,456 kW to 2,000 kW, bringing utilization from 111.4% of the 4,000 kW transformer rating down to 50%, and eliminates all three overload events that occur nightly under unmanaged charging. This is achieved while still delivering every vehicle to its 80% state-of-charge target by 07:00 IST; the discomfort term in the objective function ensures the model does not trade fleet readiness for peak reduction. The CLARABEL solver returns a globally optimal solution for the full 57,600-variable problem in approximately 3,000 milliseconds, with total end-to-end API response time, including session loading, constraint-matrix construction, and post-processing, of approximately 5,500 milliseconds on a warm server.

On the financial side, shifting the same 600-vehicle load from the 18:00-22:00 window into the 02:00-05:00 clean window reduces the DVVNL HT-2 demand charge by an estimated ₹8.60 lakh (860,000 rupees) per month, against a proposed software-as-a-service fee of ₹35,000 per month, an approximate 24.6-times return on the operator's software spend. On the carbon side, the same load shift moves 14,592 kWh of nightly charging energy from a grid segment emitting 0.771 kg CO₂/kWh to a segment emitting 0.631 kg CO₂/kWh, an estimated reduction of roughly 2,072 kg of CO₂ per night for this single depot, without requiring any change to the vehicles, chargers, or total energy consumed. Scaled across ten similarly sized depots, the same approach would correspond to an estimated 20,720 kg of CO₂ avoided per night and roughly ₹86 lakh in monthly demand-

charge savings in aggregate; at 100 depots, this scales to approximately 75,600 tonnes of CO₂ avoided annually.

A physical, scaled-down demonstrator (five relay-switched loads representing the 600-vehicle simulation at a 1:120 ratio, instrumented with a current sensor and controlled by a microcontroller acting as an OCPP dispatch endpoint) was also built to validate that the optimizer's staggered schedule corresponds to a measurable, closed-loop reduction in electrical current draw relative to simultaneous switching, providing physical corroboration of the software-level result independent of the simulation itself.

6. Discussion: A Climate Justice Framing

Climate justice scholarship distinguishes between decarbonization strategies that reduce aggregate emissions and those that do so without shifting cost or risk onto parties poorly positioned to bear it. Corporate EV fleet transitions in India illustrate this distinction concretely. A fleet operator who adopts EVs purely on emissions-reduction grounds, without addressing charging timing, may end up paying materially higher demand charges under time-of-use tariffs, a cost that, for thin-margin last-mile and staff-transport operators, is plausibly passed on to drivers through reduced compensation, to riders through fares, or absorbed as a barrier that discourages further EV adoption altogether. Uncoordinated charging can therefore create a perverse outcome in which the actors closest to the vehicles, drivers and low-margin operators, bear a cost that the climate benefit does not compensate for at their level, even as the aggregate national emissions benefit is real.

GridPilot's contribution to this problem is that it does not require operators to choose between financial and climate objectives, because the same load-shifting action, moving charging from the early-evening peak to the pre-dawn low-carbon window, simultaneously reduces DVVNL demand charges and grid carbon intensity. The joint objective function is not a coincidence of this particular

tariff and grid structure; in most Indian grids, the evening ramp that drives high carbon intensity is the same period in which utilities impose the steepest demand charges, because both reflect the same underlying scarcity of low-cost generation capacity at that hour. A scheduling system that targets one objective is likely to move substantially toward the other as a side effect, but an explicit joint formulation, as used here, ensures neither objective is sacrificed to over-optimize the other, and provides an auditable, quantified account of both savings to present to fleet operators and to regulators.

This framing also bears on distributional equity between depots. A large, well-capitalized operator can absorb transformer upgrades or demand-charge penalties that a smaller operator cannot; a software-only scheduling layer that runs on existing hardware and existing utility connections is comparatively affordable and does not require the capital expenditure that would otherwise privilege larger operators in an EV transition. At an indicative ₹35,000 per month software fee against an estimated ₹8.60 lakh in monthly savings, the tool is structured to make its adoption net-positive for the operator from the first billing cycle, which matters for adoption by smaller and mid-sized fleet operators who might otherwise be priced out of the transition's financial upside.

Finally, climate justice analysis typically asks who benefits from a decarbonization measure beyond the immediate commercial parties. Two of the revenue mechanisms considered for this system extend its climate benefit outward: participation in demand response programs under India's Central Electricity Regulatory Commission 2021 Demand Response Regulations would allow the depot to be paid by the distribution utility for load reduction during grid-stress hours, directly supporting grid reliability for the surrounding service area; and participation in India's Carbon Credit Trading Scheme (Bureau of Energy Efficiency, 2023) would allow the quantified nightly carbon reduction to be sold as credits, creating a market mechanism that rewards measured, verifiable emissions reduction rather than self-reported claims.

7. Limitations and Future Work

This study reports simulation results rather than results from an operational deployment; the reference depot, fleet composition, and building load are representative of a large Indian corporate fleet operator but are not drawn from a live production system, and the physical demonstrator built to corroborate the schedule operates at a small, scaled representation rather than at full depot scale. Three specific production gaps are acknowledged. First, the current schedule is computed once at 20:00 IST and held for the full night; a production deployment would benefit from model predictive control, re-solving the schedule at regular intervals (for example, every fifteen minutes) as actual arrivals and departures deviate from forecast, which requires a live data pipeline from physical chargers that is only available once the system is deployed against a real fleet. Second, the solver currently runs cold each night, taking approximately 3,000 milliseconds; a production system could warm-start CLARABEL from the previous night's solution and cache constraint-matrix construction, plausibly reducing total solve time to a few hundred milliseconds, though this carries no effect on the optimality or climate outcome already reported. Third, the Prophet forecasting component (0.83% validation error) is adequate at current scale but may be superseded by larger pretrained time-series foundation models as fleet size and forecasting complexity grow, a substitution that would require additional compute resources beyond what the current minimum-viable deployment uses.

The carbon-intensity figures used here are drawn from the CEA's most recent published baseline (version 21) at the time of writing; grid intensity evolves as renewable capacity is added, and the relative advantage of overnight charging will shift, in most plausible near-term scenarios narrowing somewhat as more renewable generation and storage become available around the clock, though not disappearing given the continued gap between generation profile and end-use demand timing

in the Indian grid. Finally, this paper models a single reference depot; validating the approach against multiple depots with differing building-load profiles, climate zones, and state-level tariff structures (which vary by distribution utility across India) is necessary before the aggregate national-scale estimates offered here can be treated as more than an illustrative projection.

8. Conclusion

Uncoordinated overnight charging at Indian corporate EV depots risks eroding much of the climate benefit that EV adoption is meant to deliver, while simultaneously exposing fleet operators to transformer overload and elevated demand charges under time-of-use tariffs. This paper has presented GridPilot, a convex quadratic programming framework that schedules per-vehicle charging across a 600-vehicle reference depot to jointly minimize grid carbon intensity, transformer peak load, vehicle charging discomfort, and demand-charge exposure, solved to global optimality using CVXPY and the CLARABEL solver. Simulated against Indian government fleet, grid-intensity, and tariff data, the approach reduces peak transformer load by 55.1%, eliminates nightly overload events, saves an estimated ₹8.60 lakh per month in demand charges, and avoids an estimated 2,072 kg of CO₂ per night, without altering vehicles, chargers, or total energy delivered, and while still meeting every vehicle's next-morning readiness requirement. Because the same load-shifting action addresses both the financial and the carbon objective simultaneously, and because the approach requires no new capital hardware to deploy, it offers a route to EV fleet decarbonization that does not, on its own, create new distributional costs for the operators, drivers, and communities that a corporate EV transition is intended to benefit.

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References

- Bureau of Energy Efficiency. (2023). Carbon Credit Trading Scheme (CCTS). Ministry of Power, Government of India.
- Cao, Y., Wang, H., Li, D., & Zhang, G. (2021). Smart online charging algorithm for electric vehicles via customized actor-critic learning. arXiv preprint arXiv:2106.00854.
- Central Electricity Authority. (2025). CO₂ baseline database, version 21 (FY 2024-25). Ministry of Power, Government of India.
- Central Electricity Regulatory Commission. (2021). Demand response regulations, 2021. Government of India.
- Evoltsoft Technologies. (n.d.). Smart load management and EV fleet charging software. Retrieved July 2026, from <https://www.evoltsoft.com>
- Houbbadi, A., Trigui, R., Pelissier, S., Redondo-Iglesias, E., & Bouton, T. (2019). A quadratic programming based optimisation to manage electric bus fleet charging. *International Journal of Electric and Hybrid Vehicles*, 11(4), 289-307.
- Lee, Z. J., Li, T., & Low, S. H. (2019). ACN-Data: Analysis and applications of an open EV charging dataset. *Proceedings of the Tenth ACM International Conference on Future Energy Systems (e-Energy '19)*, 139-149.
- EVreporter. (2026, January). Vahan dashboard EV market share analysis, CY2025. Ministry of Road Transport and Highways, Government of India.

- Pierrou, G., Valero-De la Flor, C., & Hug, G. (2024). Optimal EV charging scheduling at electric railway stations under peak load constraints. arXiv preprint arXiv:2404.07804.
- Uttar Pradesh Electricity Regulatory Commission. (2025, November). Order on Dakshinanchal Vidyut Vitran Nigam Limited HT-2 tariff schedule, FY 2025-26.